Summer School on Document Analysis: Open Lab

Computer
Vision Center

WEDNESDAY

27 MAY 2026

8:30	Breakfast
9:00	Hiking
12:30	Generating and Editing Visually -Rich Documents. | Silvia Cascianelli
14:00	Lunch
15:00	Learning on Graphs: GNNs in Document Analysis | Andreas Fischer
16:30	Coffee
17:00	Lab Session - GNNs
18:30	Open Lab (I): Agentic Visual Question Answering on a large Document Collection
20:30	Dinner

THURSDAY

28 MAY 2026

8:30	Breakfast
9:00	Information Retrieval | Richard Zanibbi
10:30	Coffee
11:00	Agentic: RAG, GraphRAG | Antonio Fabregat
12:30	Lab Session - Agentic GraphRAG
14:00	Lunch
15:00	Open Lab (II): Agentic Visual Question Answering on a large Document Collection
16:30	Coffee
17:00	Open Lab (III): Agentic Visual Question Answering on a large Document Collection
18:30	Open Lab: Results Presentation
20:30	Dinner

Open Lab

- Teams of around 5 people (4±1).
- At least 4h and 30 minutes to work on a Document Analysis problem.
- A 10-minute presentation sharing your results on Thursday at 18:30.
  - A demo would be nice!
- **An €850* award to the winning team sponsored by TC10/TC11**

- Evaluation
  - Science
  - Impact
  - Communication

*Provided as vouchers for online commerce.

Suggested Problem: Agentic Retrieval on a Large Document Collection

Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Document Collections

Łukasz Borchmann, Jordy Van Landeghem, Michał Turski, Shreyansh Padarha, Ryan Othniel Kearns, Adam Mahdi, Niels Rogge, Clémentine Fourrier, Siwei Han, Huaxiu Yao, Artemis Llabrés, Yiming Xu, Dimosthenis Karatzas, Hao Zhang, Anupam Datta

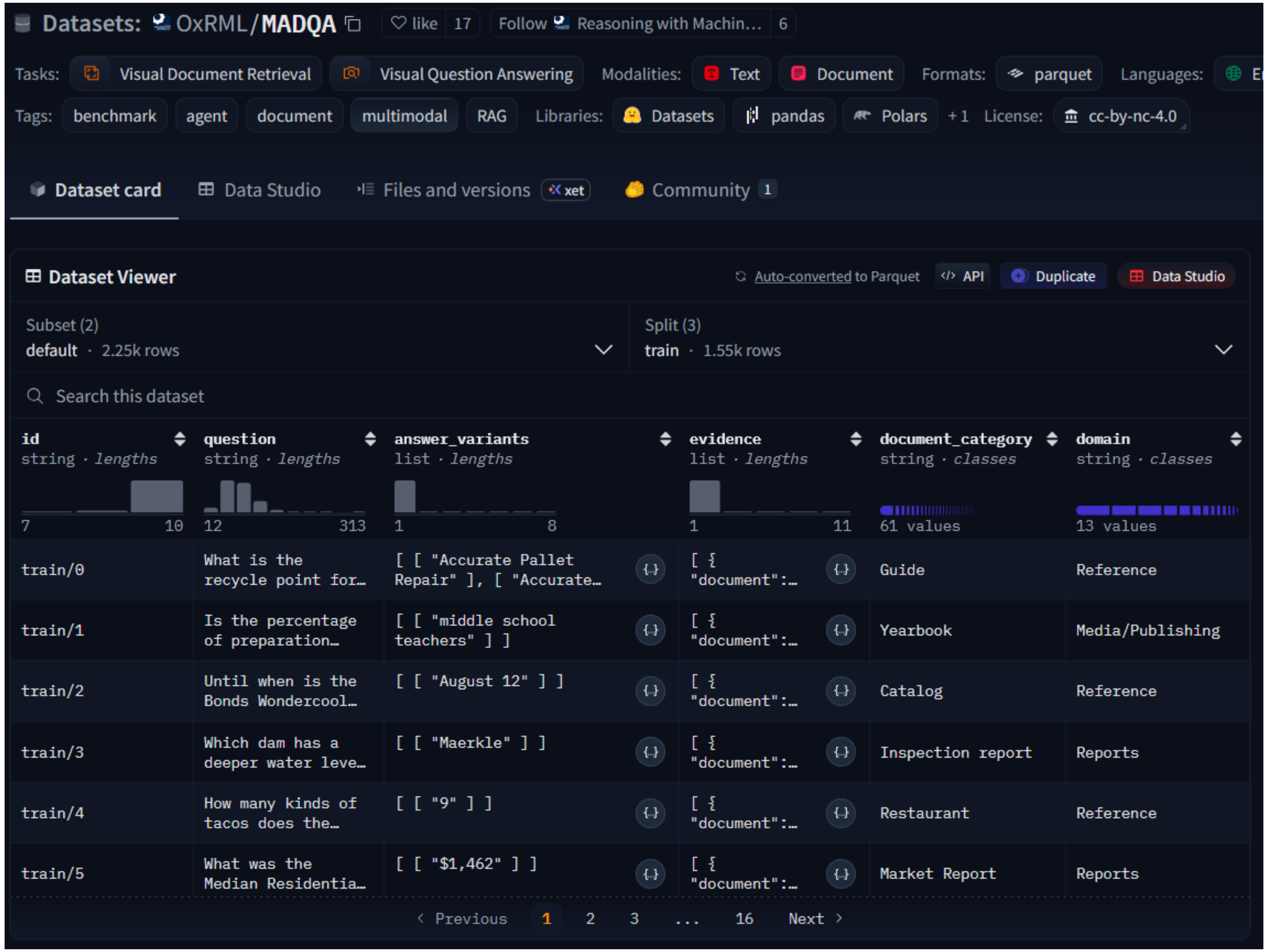
Snowflake/MADQA-Leaderboard OxRML/MADQA OxRML/MADQA

https://arxiv.org/abs/2603.12180

Suggested Problem: Agentic Retrieval on a Large Document Collection

Model / Framework	Accuracy	X-Page	X-Doc	Page F1	Doc F1	Kuiper ↓
Non-Agentic Systems						
Gemini 3 Pro File Search	**78.6** ± 2.2	**74.1** ± 3.6	**75.0** ± 3.6	**70.1** ± 2.0	**94.2** ± 1.0	–
Gemini 2.5 Flash File Search	71.8 ± 2.4	61.3 ± 4.1	73.0 ± 3.7	52.2 ± 2.2	80.9 ± 1.8	–
M3DocRAG	61.6 ± 2.6	31.0 ± 3.9	35.0 ± 4.0	68.2 ± 2.1	82.6 ± 1.7	–
GPT-5.2 (2024-08) HEAVEN	52.9 ± 2.7	38.9 ± 4.1	53.0 ± 4.2	48.4 ± 2.2	62.3 ± 2.2	–
GPT-5.2 (2025-12) File Search	50.0 ± 2.7	39.5 ± 4.1	23.0 ± 3.5	28.5 ± 2.0	68.5 ± 2.1	–
GPT-5 (2025-08) File Search	49.6 ± 2.7	36.4 ± 4.0	25.0 ± 3.6	29.3 ± 2.0	66.6 ± 2.1	–
GPT-4o (2024-08) HEAVEN	48.6 ± 2.7	32.2 ± 3.9	37.0 ± 4.0	43.2 ± 2.2	59.2 ± 2.2	–
GPT-5 Mini (2025-08) File Search	48.5 ± 2.7	32.8 ± 3.9	26.0 ± 3.7	29.0 ± 2.0	67.3 ± 2.1	–
ColBERTv2 + Llama-3.1-8B	40.2 ± 2.6	23.7 ± 3.5	26.0 ± 3.7	43.4 ± 2.2	52.0 ± 2.2	–
Agentic Systems						
Gemini 3 Pro BM25 Agent	**82.2** ± 2.0	**66.8** ± 3.9	73.0 ± 3.7	78.5 ± 1.8	90.2 ± 1.3	25.8
Claude Sonnet 4.5 (2025-09) BM25 Agent	80.6 ± 2.1	**66.8** ± 3.9	**82.0** ± 3.2	**79.1** ± 1.8	**93.0** ± 1.1	35.1
GPT-5 (2025-08) BM25 Agent	77.7 ± 2.2	60.1 ± 4.1	74.0 ± 3.7	74.2 ± 2.0	86.5 ± 1.5	52.6
Gemini 3 Pro RLM	73.8 ± 2.3	**66.8** ± 3.9	66.0 ± 3.9	69.1 ± 2.1	89.8 ± 1.4	**22.9**
Claude Agent Semtools	72.6 ± 2.4	62.0 ± 4.0	60.0 ± 4.1	51.1 ± 2.2	89.5 ± 1.4	37.9
Claude 4.5 Sonnet (2025-09) RLM	70.5 ± 2.4	65.0 ± 4.0	69.0 ± 3.9	66.5 ± 2.1	88.9 ± 1.5	42.3
Claude Haiku 4.5 (2025-10) BM25 Agent	68.2 ± 2.5	48.0 ± 4.2	65.0 ± 4.0	72.0 ± 2.0	88.2 ± 1.4	50.7
GPT-5.2 (2025-12) BM25 Agent	67.8 ± 2.5	51.6 ± 4.2	55.0 ± 4.1	67.6 ± 2.1	83.7 ± 1.7	64.8
GPT-5 Mini (2025-08) BM25 Agent	66.9 ± 2.5	48.0 ± 4.2	48.0 ± 4.2	67.6 ± 2.1	82.4 ± 1.7	73.2
GLM-4.6V BM25 Agent	66.1 ± 2.5	37.1 ± 4.0	70.0 ± 3.8	65.9 ± 2.1	86.6 ± 1.5	51.4
GPT-5.2 (2025-12) RLM	64.2 ± 2.6	55.3 ± 4.1	56.0 ± 4.1	67.6 ± 2.1	83.7 ± 1.7	30.0
MDocAgent	63.8 ± 2.6	37.1 ± 4.0	41.0 ± 4.1	67.1 ± 2.1	82.1 ± 1.7	27.1
Qwen3-VL (235B-A22B-Thinking) BM25 Agent	60.3 ± 2.6	36.4 ± 4.0	55.0 ± 4.1	58.6 ± 2.2	80.5 ± 1.8	36.6
GPT-4.1 (2025-04) BM25 Agent	60.0 ± 2.6	42.5 ± 4.1	44.0 ± 4.1	64.1 ± 2.1	82.8 ± 1.7	43.2
Gemini 2.5 Flash BM25 Agent	58.5 ± 2.6	30.4 ± 3.8	56.0 ± 4.1	61.0 ± 2.2	78.8 ± 1.8	46.5
GPT-5 Nano (2025-08) BM25 Agent	58.2 ± 2.6	32.8 ± 3.9	35.0 ± 4.0	60.9 ± 2.2	82.2 ± 1.7	49.8
Qwen3-VL (8B-Thinking) BM25 Agent	47.3 ± 2.7	17.0 ± 3.1	44.0 ± 4.1	47.6 ± 2.2	69.4 ± 2.1	50.2
GLM-4.6V Flash BM25 Agent	46.0 ± 2.7	18.2 ± 3.2	47.0 ± 4.2	28.9 ± 2.0	51.5 ± 2.2	27.5
GPT-4.1 Nano (2025-04) BM25 Agent	19.5 ± 2.1	6.1 ± 2.0	9.0 ± 2.4	27.6 ± 2.0	40.2 ± 2.2	28.6
Human Performance						
Human Oracle Retriever	**99.4** ± 0.4	**100.0**	**98.0** ± 1.2	–	–	–
Human BM25 Agent	82.2 ± 2.0	79.6 ± 3.4	72.0 ± 3.7	79.3 ± 1.8	93.4 ± 1.1	**14.6**

Suggested Problem: Agentic Retrieval on a Large Document Collection



Datasets: OxRML/MADQA like 17 Follow Reasoning with Machin… 6

Tasks: Visual Document Retrieval Visual Question Answering Modalities: Text Document Formats: parquet Languages:

Tags: benchmark agent document multimodal RAG Libraries: Datasets pandas Polars +1 License: cc-by-nc-4.0

Dataset card Data Studio Files and versions xet Community 1

Dataset Viewer Auto-converted to Parquet API Duplicate Data Studio

Subset (2) default · 2.25k rows

Split (3) train · 1.55k rows

Search this dataset

id string · lengths 7 – 10	question string · lengths 12 – 313	answer_variants list · lengths 1 – 8	evidence list · lengths 1 – 11	document_category string · classes 61 values	domain string · classes 13 values
train/0	What is the recycle point for…	[["Accurate Pallet Repair"], ["Accurate…	[{ "document":…	Guide	Reference
train/1	Is the percentage of preparation…	[["middle school teachers"]]	[{ "document":…	Yearbook	Media/Publishing
train/2	Until when is the Bonds Wondercool…	[["August 12"]]	[{ "document":…	Catalog	Reference
train/3	Which dam has a deeper water leve…	[["Maerkle"]]	[{ "document":…	Inspection report	Reports
train/4	How many kinds of tacos does the…	[["9"]]	[{ "document":…	Restaurant	Reference
train/5	What was the Median Residentia…	[["$1,462"]]	[{ "document":…	Market Report	Reports

‹ Previous 1 2 3 ... 16 Next ›

https://huggingface.co/datasets/OxRML/MADQA

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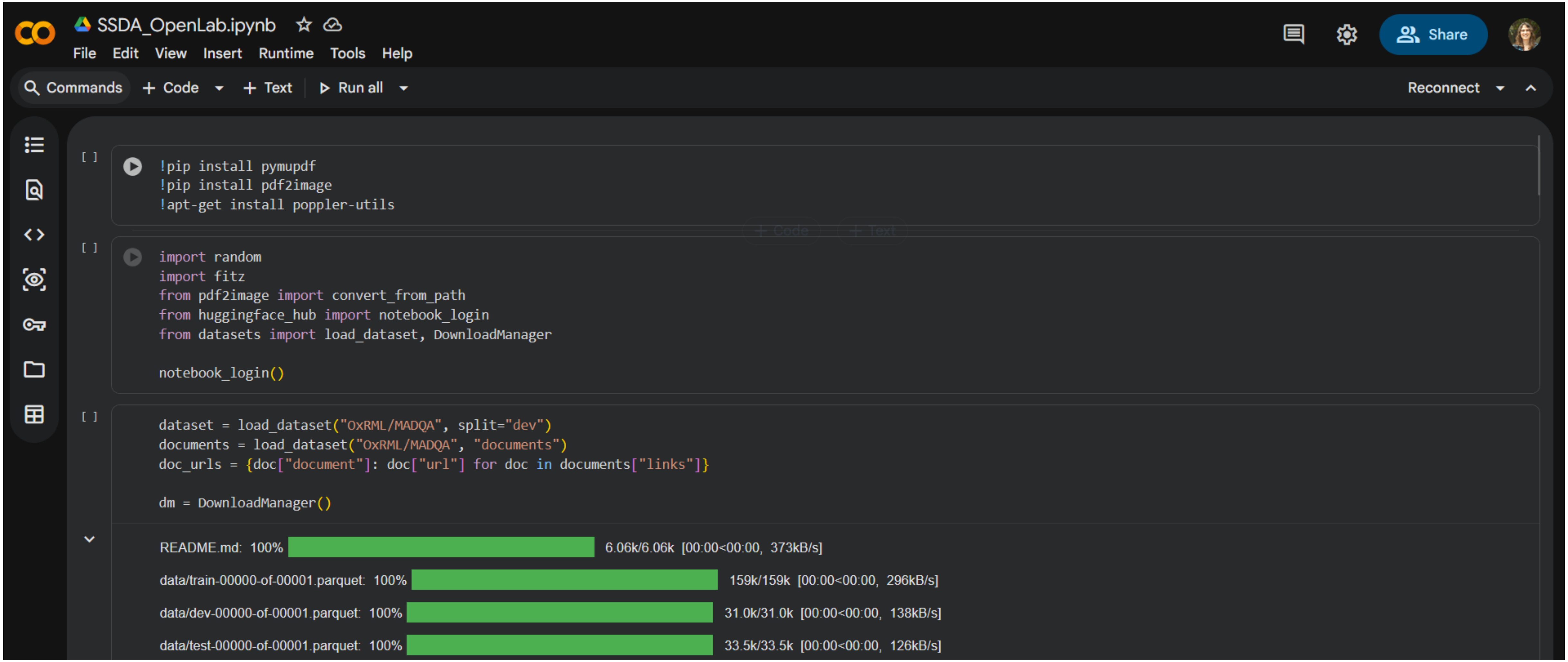

https://colab.research.google.com/drive/1fN-4yDuJ5EY3_EyAoJyqzZnwlv8KZ64j

Time to work!

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